AI Powered Cloud Operations Platform

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AGENDA

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AI-Ops Ninjas Team!!!

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Problem Statement

Problem

Cloud environments generate **1,000s of alerts daily**
(CPU spikes, failures, security, cost anomalies)

Current State

Manual DevOps/SRE triage →

- **MTTR in hours** (delayed resolution) → $100k per incident (due to prolonged outages)
- **80% alerts are noise** (alert fatigue) → $1M+ per year (due to wasted engineering time)
- **High risk of human errors → non-compliance exposure / security breach**

Solution

We need **autonomous cloud operations** that can **detect, analyze, and remediate issues automatically**

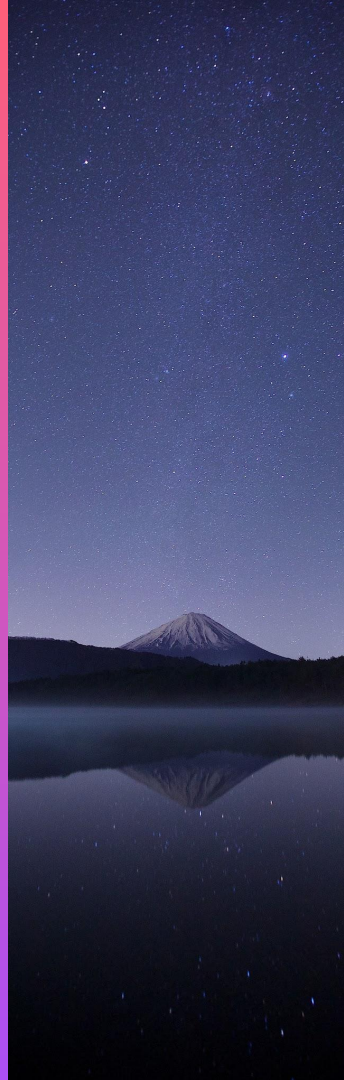
The Solution

Build an **AI-powered Cloud Operations Platform** that can:

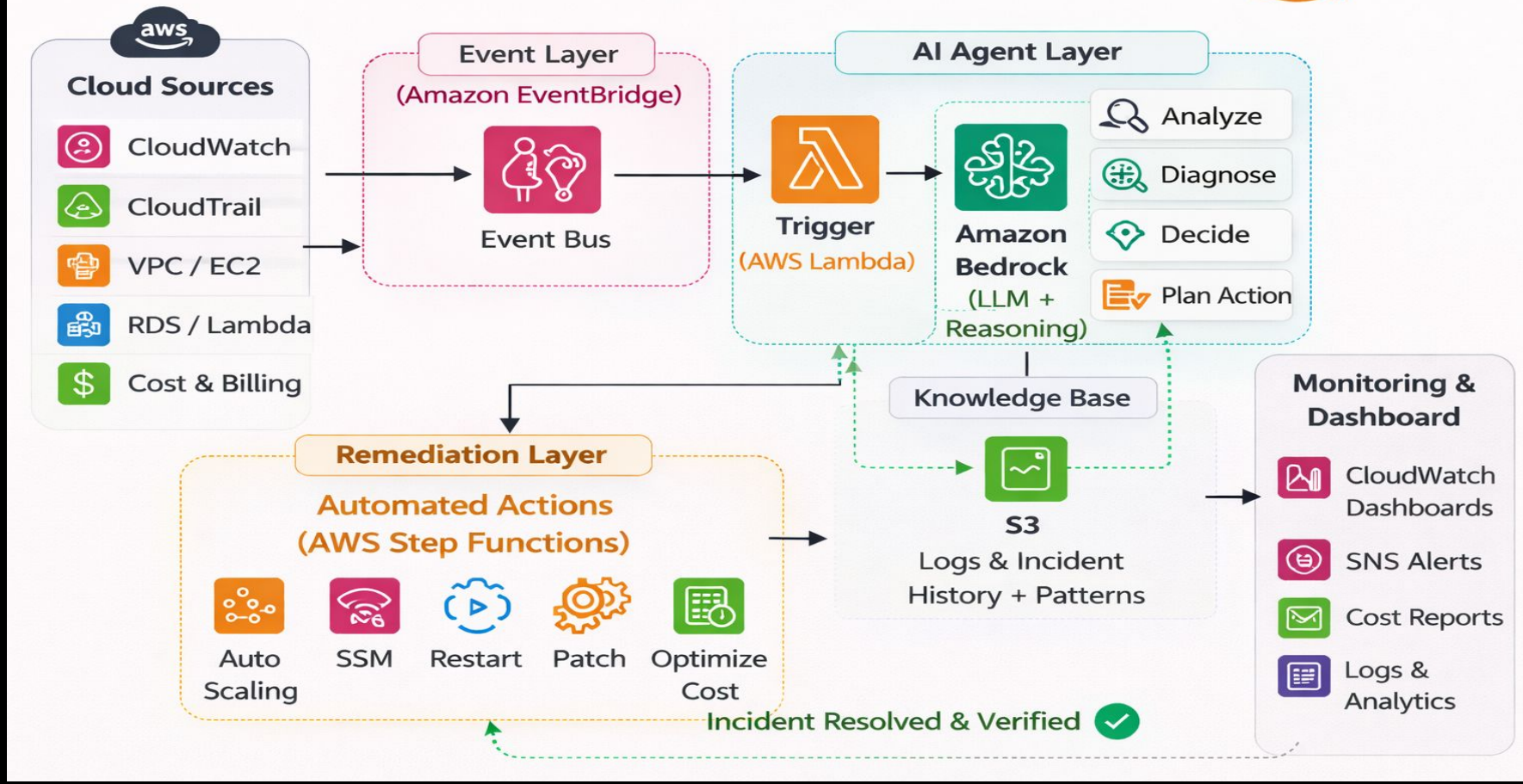
- Detect operational events in real-time/near real-time
- Analyze root cause using AI reasoning
- Decide remediation action
- Execute automated fix
- Notify engineers
- Reduce MTTR by **40–50%**

This creates a **self-healing cloud system and generic solution which can be used across all organizations.**

https://github.com/rkumarus040599/sa-bcamp-hackathon-2026



Agentic AI for Autonomous Cloud Operations on AWS



More about the architecture

Follows the principles of Well Architected Framework

→ Cost effective (All serverless services)

→ Secure

→ Highly available

→ Scalable

Demo - v2

https://github.com/rkumarus040599/sa-bcamp-hackathon-2026

Demo

CloudWatch Alarm
↓
EventBridge
↓
Event Router Lambda
↓
AI Agent Layer(Amazon Bedrock RAG) ←-> Knowledge Base
↓
AWS Step Functions (Remediation Layer) ←> S3 (augmenting historical data)
↓
Monitoring & Dashboard (for Visualization)

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Visual view of the Demo

https://github.com/rkumarus040599/sa-bcamp-hackathon-2026

Next Steps

AI - Ops: Phase 2 and beyond

→ Support additional sources (ex: VPC Flow Logs/DNS Logs)
→ Additional metric triggers (ex: 5-Tuples (SIP/DIP), …)
→ Additional historical data (ex: 60/90/180 day historical events)
→ Customized Dashboard (Cloud admin / Cloud Ops engineer)
→ Additional Tool capabilities for LLM
→ Additional Remediation capabilities (ex: adding/removing EC2 via ASG with reasoning capabilities / isolating the EC2 instance for forensic analysis).

THANK YOU

Any questions?